1.2 The Sociophonetics of /s/

By varying the tongue position with which an /s/ is pronounced, a speaker can vary the frequencies (or pitches) of the sound produced; when the tongue position is fronter (closer to the teeth) the resulting /s/ has primarily higher frequencies, when the tongue position is backer (further from the teeth), the resulting /s/ has primarily lower frequencies (Fuchs and Toda 2010). A common measure of this frontness is the spectral center of gravity (CoG), which is the mean of all of the frequencies of an /s/ sound at a given moment in time, weighted for their amplitudes (volumes), measured in Hz. It is important to note at the outset that gendered differences in /s/ frontness do not arise from any anatomical differences associated with the sex binary, unlike other gender-correlated sites of variation such as overall vocal pitch; all people with typical vocal tracts have the anatomical equipment to produce roughly the same range of /s/ pronunciations (Zimman 2017c; 2021b).

Research on white, English-speaking, cisgender heterosexual men and women in North America – which did not include non-White, non-English-speaking, or queer speakers and listeners, a point to which I will return – find robust correlations between speaker gender and /s/ pronunciation. Flipsen et al. 1999 review 21 such studies, finding that women's /s/ pronunciations are consistently fronter than men's. Listeners in this population are aware of these correlations, and judge fronter /s/ to sound feminine, and backer /s/ to sound masculine. Strand (1999) finds that when asked to to categorize audio stimuli on a spectrum of /s/ frontness as either /s/ or /ʃ/[1] (/ʃ/ is similar to a very backed /s/), they place the boundary higher when told the speaker of the audio clip is a woman than when they are told it is a man, thus exhibiting a link between /s/ frontness and percieved gender. When listeners in this population percieve a mismatch in /s/ pronunciation and the percieved gender of a voice, they judge that voice to sound gay or lesbian. A number of studies have focused on this dynamic in perception of men's voices (eg. Levon 2014; Campbell-Kibler 2011). Munson, Jefferson, and McDonald (2006), replicates Strand's experiment, finding when listeners perceive a speaker to be lesbian sounding, they place the boundary between /s/ or /ʃ/ lower, thus expecting lesbians to back /s/.

Calder (2021) points out that these earlier correlational studies often interpret indexicalities, but fail to situate these indexicalities as specific to the populations in which they are found. It is insufficient to claim that fronter /s/ pronunciations index femininity and gayness and backer /s/ pronunciations index masculinity and lesbianness without specifying for whom these indexicalities hold. These indexical links are made specifically by cisgender heterosexual white listeners, who take the behavior of their own population as the norm, and interpret others through theis norm (an act of indexical presumption). Calder (2021) thus considers indexical links from /s/ frontness to femininity and gayness to be gender norms within the hegemonic indexical field for /s/, rather than neutral and universalizable patterns.

Calder (2021) proceeds to discuss two examples of research on outside of this

[1] /ʃ/ represents the English 'sh' sound, as in the beginning of 'shirt' or the middle of 'station'

hegemonic indexical field, one investigating Black American speakers' /s/ pronunciations, and one investigating queer nonbinary drag queens'. In the first of these studies, J. Calder and King (2022) measure the speech of find that cisgender heterosexual Black speakers in Bakersfield, California /s/ pronunciations were not differentiated by speakers' gender. They highlight that white normativity does not apply here; the men and women who participated in this study considered themselves normatively gendered cisgender heterosexuals, and cannot be said to be improperly indexing their genders, or indexing queerness. Within the indexical field of speakers in this community, Calder and King argue /s/ comes to be an index of white femininity/masculinity, and thus irrelevant to (or even to be avoided by) black speakers.

Looking at racialized aspects of /s/ in perception, Maegaard and Pharao (2015) and Pharao et al. (2014) present a similar finding in a different context. In Copenhagen Dutch, /s/ has been shown to pattern and be percieved similarly to the North American white cisgender heterosexual English-speaking population. Maegaard and Pharao find that young listeners (both white and nonwhite) in only hear fronted /s/ as feminine and gay when it is within the context of a prosodic pattern associated with middle-class whiteness, 'modern' standard Copenhagen Danish speech; in the context of a prosodic pattern associated with lower-class and immigrants, 'street' Danish, listeners instead hear fronted /s/ as associated with 'gangster'-ness and masculinity. Although Maegaard and Pharao do not apply the concept of Indexical fields to their work, it can be said that /s/ has two indexical potentials within the indexical field of these listeners, and that other indexical behavior, in this case the prosodic pattern, determines which set of indexicalities speakers take up – if the speaker sounds white with the 'modern' prosodic pattern, /s/ is interpreted through white gender norms, if the speaker sounds nonwhite with the 'street' prosodic pattern, /s/ is intepreted differently.

From these examples, two points are clear. First, local or marginalized communities' indexical fields are not simply independent of hegemonic meanings, but are instead interlinked with them, qualifying and reinterpreting hegemonic meanings. Second, variables like /s/ do not work on their own, but only come to be interpretable through a broader semiotic context; listeners in Maegaard and Pharao's studies could only interpret /s/ through its combination with prosodic types. In the second of the studies mentioned above, Calder (2019a) applies the theories of indexical orders developed by Silverstein (2023; 2003), and fractal recursivity developed by Irvine and Gal (2000) and Gal and Irvine (1995) in order to investigate this process.

As linguistic forms are construed as indexical signs and gain meaning within a particular indexical field, it is possible that new meanings will arise which build off of and are informed by the original meaning. This can take place both within a single indexical field, and cross-cutting the overlaps of multiple. The development of an indexical link between a linguistic form and a social meaning always already emerges informed by an uncountable chain of previous indexicalities and meaning-making, and for this reason is said to exist at the nth indexical order (Silverstein 2003). In the case of /s/, Calder understands the indexical link between the gender

binary and /s/ frontness to exist at the nth order within the hegemonic indexical field. When cisgender heterosexual white listeners' perception of speakers' gender does not match with their perception of that speaker's /s/ pronunciation, this recontextualization of /s/ usage forms the $n+1$st indexical order, in which a new interpretation must emerge. Here, the meaning of /s/ comes to be qualified by other construals of meaning, in that a listener's perception of a speaker's gender must be based upon other indexical conjecturing. That cisgender heterosexual white listeners construe this percieved mismatch in speaker gender and /s/ pronunciation as as gay/lesbianness evidences an erasure of transgender, nonbinary, and other queer identities; gender is assumed to be fixed, and all gender nonconformance is assumed to be gay or lesbian. Without these cisgender assumptions, the $n+1$st indexical meaning of a mismatch of percieved speaker gender and /s/ would simply be gender nonconformance, or perhaps that the speaker's gender was percieved incorrectly. With these assumptions, it comes to be gay/lesbianness. It is important to note that although the nth and $n+1$st indexicalities of /s/ are situated at different orders, they nonetheless inform and construct each other, and do not form a chronological order despite their ordered structure.

Within the San Francisco drag communities Calder (2019a) studies, an additional indexical order of /s/ is forged at the $n+1+1$nd level which is local to these communities' indexical fields, escaping the hegemonic field. A drag community Calder studies in the South of Market (SoMa) neighborhood of San Fransisco figures itself as more radically queer than another drag community in the Castro neighborhood. This is reflected by the differences in identities between these communities; SoMa queens generally identify as beyond the gender binary, many of them nonbinary or transgender, whereas the queens of the Castro neighborhood primarily identify as cisgender gay men. Calder finds that SoMa queens front /s/ more than Castro queens, and that this forges a new local indexicality to 'fierceness', anti-normativity, and and in-group status at the at the $n+1+1$nd level. Their diagram of this structure of meanings is included in figure 1.1. This emerges as an example of indexical orders cross-cutting indexical fields, where meanings in the hegemonic cisgender heterosexual white indexical field came to inform the creation of new meanings in the local, alternative indexical field.

As mentioned above, within Calder's model fronting /s/ emerges as deviant from gender norms only in those whose gender is understood as 'male' within the cisnormative conception of gender as bodily characteristics. It is perhaps surprising, then, that nonbinary and transfeminine speakers in the SoMa drag community, who would not like to be percieved as men, front /s/ exactly for the purpose of gender defiance, rather than gender conformance. It is helpful here to introduce a theoretical tool Calder employs to understand the interlinking of identity categories at these different levels of sex at birth, gender identity, sexuality, and trans/queerness– the semiotic process of fractal recursivity, as theorized by Irvine and Gal (2000) and Gal and Irvine (1995).

In fractal recursitivity, a semiotic opposition on one level becomes nested into itself, and this process can happen any number of times (Irvine and Gal 2000; Gal and Irvine 1995). In moving from the nth indexical order of /s/ to the $n+1$st (both

of which are within the hegemonic indexical field), the gender binary is nested into itself, creating a metaphorical structure where men and women are split into binary gender oppositions. This can be understood as distinguishing some notion of 'men who are men' from 'men who are women', and 'women who are women' from 'women who are men'. These become the metaphorical representation of gay men and lesbian women, since members of these groups are still fundamentally understood as types of men and types of women, relying on the aforementioned erasure of trans people and those beyond the binary. Is it then the case that by situating the distinction between cisgender gay Castro queens and nonbinary and transgender queer SoMa queens as nesting into the 'men who are women' branch, Calder as the analyst, and possibly the SoMa queens themselves, tacitly consider these queens as nonetheless a 'type of man/male'? I argue it is not. Instead, in my reading of Calder's work, SoMa queens reinterpret at the $n + 1 + 1$st level such that the fractal-recursive binary structure is broken; instead, the opposition is cisgender gay men vs. non-cisgender queer people, or as Calder articulates it, gay vs. queer. This binary still fits with the base level of /s/ indexicality, which is always present; it is not difficult for SoMa queens to figure gay men as less feminine and less subversive than queer non-cisgender people. From here, the indexicality of /s/ frontness as subversive queer non-cis femininity can be less problematically interpreted.

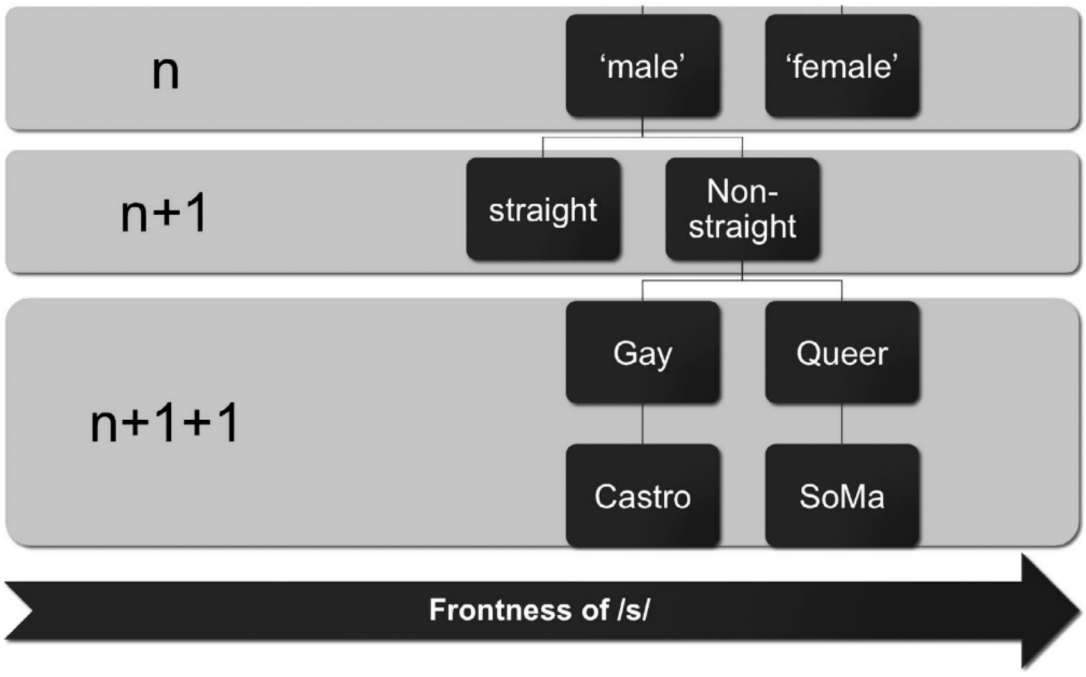


Figure 1.1: Diagram from Calder (2019a, p. 340) illustrating indexical orders of /s/ frontness and corresponding fractal recursive social oppositions across two San Fransisco drag communities.

A. J. Steele (2022) provides another example of /s/ being reinterpreted by those

outside of the gender binary. They investigate a racially-diverse community of Columbus, Ohio nonbinary individuals, finding that for Black speakers, fronted /s/ seemed to have the opposite indexical meaning, correlating positively with speakers' self-rating of their masculinity. For white speakers in this community, there was no correlation between /s/ pronunciations and masculinity ratings. This leads Steele to conclude that Black speakers subvert /s/ meanings not only to create a nonbinary identity and differentiate themselvelves from a cisgender norm, but also to repurpose and subvert a white norm. For white speakers, this dimension is not present, so the behavior is different. These patterns suggest that nonbinary speakers in this community forge further indexical orders which allow them to negotiate their specific racial and gender positions against white and cisgender normativities.

The studies discussed above illustrate the ability of speakers outside of cisgender heterosexual white populations to construct alternative and subversive meaning structures for /s/ which allow an interpretation of their unique speech practices as innovative and subversive, rather than inadequate and deviant. However, due to the dominance of the hegemonic indexical field, and the practicalities of social life, it is nonetheless the case that marginalized speakers outside of cisgender heterosexual white populations have to deal with these norms. Although it is important to note that these all speakers are socialized to believe in these norms, even if these norms will marginalize them, I focus on the most certain case, in which marginalized speakers interact with those these norms directly center and privilige.

That the hegemonic structure of indexicality for /s/ particularly excludes trans speakers is attested to by the findings of Jessee and Calder (2025). They investigate cisgender perceptions of transgender /s/ by replicating Strand (1999), comparing the placement of the boundary between /s/ and /ʃ/ when the perceiver is told the speaker is trans vs. not told they are trans (and thus is likely to assume that they are cis). They find that unlike in Munson, Jefferson, and McDonald (2006), where cisgender heterosexual speakers expect cisgender gay men and lesbians not to conform with the hegemonically normative /s/ pronunciation for their gender, transgender speakers are expected to hyper-conform– trans women are expected to front /s/ more than cis women by all listeners, and listeners who do not know a trans person also expect trans men to back /s/ more than cis men. Although indexicalities certainly would have changed in the 20 years between these two studies, it is nonetheless surprising that cisgender heterosexual white listener holds different expectations and perceptions for trans people than for cisgender gay men and lesbians, especially since it is counter to trans speakers' actual behavior; Podesva and Van Hofwegen (2015), Hazenberg (2012), and Zimman (2021b) find that trans speakers' /s/ pronunciations tend to be closer to the center of the spectrum, not as strongly gender-differentiated than cis speakers'.

Jessee and Calder (2025) analyze this through the lens of the cisgender listening subject, seeing that rather than basing perceptions on transgender speakers' actual speech, "the cisgender listening subject, which expects voices to conform to normative standards of gender appropriateness, places even stronger standards of

gender normativity on trans speakers than it does on cis speakers." (9-10) Here, the cisgender listening subject will only accept an idealized highly-conforming trans person, and construes real trans speakers as inadequately performing their genders. Because we do not live up to this idealized standard, cis percievers will misgender trans speakers, inevitably hearing them as the gender they were assigned at birth, rather than their actual gender. In combination with the $n + 1$st order of meanings for /s/, this means that attempts to move towards fulfillment of hegemonic gendered /s/ pronunciations are likely to be misrecognized as indexical of gay and lesbianness, rather than transfemininity and transmasculinity respectively. Trans speakers are 'damned if they do, damned if they don't' – if a trans woman does not front /s/, her transness may be questioned, and she may be heard as a man; if she fronts /s/, she may be construed only as a gay man, not a trans woman. In this difficult indexical situation, in which a trans speaker knows their cisgender listener may subject them to this hegemonic indexical field which excludes them, there are two primary options; defend oneself by moving towards adherence to these norms, or refuse to do so, resisting these norms and risking negative judgements. As mentioned above, transfeminine speakers studied by Heffernan (2023) consider both of these options, acknowledging the safety and possible gender-affirmation in taking up hegemonically normative femininity, while simultaneously not wishing to judge themselves by cisgender standards. Drawing upon all of this prior work on indexicalities of /s/, this thesis investigates whether transfeminine speakers shift towards hegemonically-feminine fronter /s/ pronunciations with cis listeners, and what motivations might determine the choice to shift or not shift in view

That individuals may change their /s/ pronunciations within interaction for gender-presentational and interactional purposes has been attested by Benesch (2022), Mendoza (2024), Saigusa (2016), Holmes-Elliott and Levon (2017), Levon and Holmes-Elliott (2013), Parnell-Mooney (2019), Calder (2019b), and Zimman (2017c). Holmes-Elliott and Levon (2017) and Levon and Holmes-Elliott (2013), in a study of within-speaker variation in /s/ pronunciation in British reality television shows, find that in one lower-class show, cisgender heterosexual women use backer /s/ in conversation with other women, whereas in a higher-class show, cisgender heterosexual women use fronter /s/ in this context. In order to explain these contradictory results, they conclude that the interlocutor's gender may not be the best predictor, as in both shows women back /s/ in speech activities considered more threatening. As discussed below in the methods chapter, the recorded conversations I conducted attempted to control for this stance-based shifting through the structure of two collaborative-valenced speech activities.

Moving to investigate /s/ shifting in queer speakers, Calder (2019b) finds that as SoMa drag queens get into their drag, visually transforming into their drag persona, their /s/ pronunciations get progressively fronter to index their change in persona. Benesch (2022) finds that among genderfluid people, who experience different genders at different times, shifts in /s/ index these shifts in gender, however they do not necessarily correlate with typical scales of gender such as masculinity vs. femininity, but rather with personal, alternative scales such as 'flannel' vs. 'glitter'. Mendoza (2024) presents a case study of a drag queen in Wynwood, Mi-

ami, who uses fronter /s/ pronunciations when discussing the aspects of their drag taken from white, higher-class femininity, but uses backer /s/ when discussing to their more anti-normative *chonga* drag persona. *chonga* is a queer subversive Latinx femininity which may look like "unshaved body hair and beards, as well as hoop earrings and dramatic lip liner" (307). The polyvalence of this queen's drag persona is indexed through these changes in /s/. In a similar way to the findings of Calder (2019b) and A. J. Steele (2022), this queen reinterprets the indexicality of fronted /s/, using it not to index of femininity in general, but rather to index a specific drag satire of white cisgender heterosexual higher-class femininity. In this way, meanings evolve outside of the hegemonic field.

I choose to focus on /s/ for my study, rather than other linguistic forms bearing social indexicalities, because of the theoretical depth of the research that has already been conducted on /s/; as reviewed above, this research has captured a nuanced and detailed view of the hegemonic indexical field of /s/, how it is marginalizing, and how those who are marginalized by it reinterpret and construct its meanings subversively within their communities. However, in the /s/ literature, as well as in sociolinguistic research more broadly, a gap I address is the dearth of research looking specifically at situations of indexical constraint, analyzing how marginalized speakers navigate listeners' indexical fields which marginalize them. Comparing transfeminine individuals' /s/ pronunciations in conversations with another transfem, where they can adopt their own indexicalities more freely, against their /s/ pronunciations in conversations with a cis woman, is illuminating here. More broadly, this research also addresses the absence of within-speaker research on transfeminine speakers– although there have been of studies of within-speaker variation in transmasculine people (eg. Zimman 2017c; Parnell-Mooney 2019), nonbinary people (Gratton 2016), and drag queens (Calder 2019b; Mendoza 2024), there have been no such studies focused on how transfeminine speakers speak differently in differing situations. Of specific interest is the question of individuals' navigation of the $n+1$st indexical order of /s/ frontness linked to gay masculinity; whether transfeminine speakers understand this to be a possible judgement of the cisgender listening subject, and whether this is taken into account in the choice to shift /s/. In sum, this thesis pursues the research question:

> Do transfeminine speakers shift towards fronter /s/ pronunciations with a cis woman listener versus with a trans woman listener? What factors determine this choice?

Based on the literature reviewed, I predict:

1. Speakers who express greater feelings of danger, discomfort, dysphoria, or desire to pass around cisgender people are more likely to shift;
2. Speakers who differentiate themselves from or hold attitudes of resistance against hegemonic femininity are less likely to shift;
3. and Speakers who express concern with gay-male misrecognition are less likely to shift.

These predictions are based on the literature reviewed above, primarily the structure of the indexicality /s/ within the hegemonic indexical field it has established. Because of transfeminine people's awareness of the cisgender listening subject's application of hegemonic norms, I expect that transfeminine speakers' decision making around /s/ shifting will be based upon their relationship with the meanings of /s/ in the hegemonic field, as well as their relationship with hegemonic feminine norms itself; if speakers wish to distance themselves from hegemonic femininity, they will be more likely to disregard the judgements of the cisgender listening subject and refuse to take up a fronter /s/ with a cisgender listener. However, if speakers feel more threatened than defiant, feeling greater pressure to conform to hegemonic femininity, they are likely to shift towards fronter /s/ pronunciations in self-defense and to match up to the hegemonic standards. Yet, if speakers are concerned with being mistaken for a gay man via their voice, they may feel that even if they would be able to benefit from indexing hegemonic femininity, that /s/ is not an available resource through which to do so.